UNIVERSITY OF GHANA

DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

STAT 112 : PROBLEM SHEET TWO

1. Consider a discrete random variable X with pmf given by

$$P(X=x)=\begin{cases} \frac{1}{3}, & x=-2,0,2 \\ 0, & \text{elsewhere} \end{cases}$$

Show that $P(X=x)$ is legitimate and find the mean and variance of X.

2. The probability distribution of a discrete random variable X is shown below:

X	0	1	2	3
$P(X=x)$	c	c^2	c^2+c	$3c^2+2c$

Where c is a constant. Calculate:
(i)c (ii)$P(0 \leq X \leq 2)$ (iii)$P(X>1)$ (iv)$E(X)$ (v)$Var(X)$

3. Let X be a discrete random variable with probability function

$$f(x)=\begin{cases} \frac{1}{3}, & x=3 \\ \frac{2}{3}, & x=9 \end{cases}$$

Find $E(3-2X)$ and $E(X+4X^2)$

4. Let X be a random variable with $E(X)=1$ and $E(X(X-1))=4$. Find $Var(3X-2)$.

5. Let X be a continuous random variable with pdf given by

$$f(x)=\begin{cases} \frac{1}{4}(1-x^2), & \text{for } -1<x<1 \\ 0, & \text{elsewhere} \end{cases}$$

Let $h(x)=2-3x$. Determine the $Var(h(x))$.

6. A random variable X had pdf given by

$$f(x)=\begin{cases} c\sqrt{x}, & 0<x<1, \\ 0, & \text{elsewhere} \end{cases}$$

(a) Find the value of the constant c.

(b) Calculate $P(X<\frac{1}{4})$

7. A random variable X had pdf given by

$$f(x) = \begin{cases} \frac{1}{10}, & 0 < x < 10, \\ 0, & \text{elsewhere} \end{cases}$$

Find:
(a) $P(X > 8|X > 5)$ (b)$P(X > 7|X < 9)$

8. The discrete random variable X assumes values 0,1 and 10 only. Given that

$$E(X) = 2 \text{ and } Var(X) = 13$$

Determine the cumulative distribution function of X.

9. The probability distribution functon of a continuous random variable X is given by

$$f(x) = \begin{cases} 12x^2(1-x), & \text{for } 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

(i) Given that the random variable $Y = 3X - 4$, find the mean and variance of Y.

(ii) Find the probability that X lies between the mean and the mode.

10. A discrete random variable has probability mass function

$$P(X = x) = \begin{cases} a(3-x), & x = 0, 1, 2 \\ b, & x = 3 \\ 0, & \text{elsewhere} \end{cases}$$

Given that $E(X) = 1.6$, find

(i) the values of a and b.

(ii)$E(3X - 2)$ and show that $Var(X) = 1.64$.

11. The following table gives the cumulative distribution function(c.d.f) of a discrete random variable,X.

X	1	2	4	5	6	8
$P(X \le x)$	0.1	0.3	0.6	0.85	0.9	1.0

(i) Derive the probability mass function(p.m.f) of X.

(ii) Find $P(X > 5)$ and $P(3 < X \le 6)$

(b) Suppose that the random variable Y is equl to the number of hits obtained by a certain baseball player in his next 3 bats. Thus,Y takes on the values 0,1,2,3. If $P(Y = 0) = 0.35$ and $P(Y = i) = 2iP(Y = i + 1)$ for $i = 1, 2$. Find $E(Y)$.

12. Suppose the random variable X has the probability density function (p.d.f) given as

$$f(x) = \begin{cases} c(1 - x^2), & \text{for } -1 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

for some constant c.

(i) What is the value of c?

(ii) Derive the cumulative distribution function of $X, F(X)$.

(iii) Hence or otherwise, find $P(X > 0.7)$.

13. (a)State the characteristics of a binomial experiment.

(b)The Ghana Health Service claims that only 20% of young adults can pass its physical fitness test. Suppose nine young adults are randomly selected and each is given the fitness test.

i. If W is the random variable that denotes the number of the nine young adults that pass the test, what is the name given to W?
ii. What is the expected number of young adults who pass the test?
iii. What is the probability that at least one of the young adults passes the test?

14. If X is a binomial random variable with expected value 6 and variance 2.4. Find the probability of X=5.

15. Of the items produced on a certain assembly line, 10% are defective. Suppose that 25 items are selected at random,

i. Find the probability distribution of the random variable X, where X denotes the number of defective items in the sample.
ii. Find the probability that at least two items are defective.

16. A box contains 10 white and 20 black marbles. One selects at random 10 marbles with replacement. Compute the probability of obtaining at least 2 white marbles.

17. GLOBAL, is an old-established firm. The firm wishes to go public, by trading on the Ghana Stock Exchange. It has been reported that eighty percent of the stockbroking firms are recommending the issue to their clients. Five(5) stock broking firms have been selected at random to know their view on GLOBAL's initial offer. Find the Probability that at least 3 of them are recommending the purchase of this issue to their clients.

18. The proportion of defective items in a certain manufacturing process is 0.20. If 10 items are selected from the process, Find

    a. The expected number of good items

    b. Probability that none is defective

    c. Probability that at least 3 are defective.

19. There are five old ATM machines at Smart Bank. All probability that an ATM breaks down in the course of a day is 0.2. If they are all in working order at the beginning a day, and breakdowns are independent of each other, find the probability that during the day;

    i. there are no breakdowns

    ii. four machines break down

    iii. all machines break down

    iv. at least three machines break down

20. Given that $X \sim B(n, p)$, Show that

    (a) $E(X(X-1)) = n(n-1)p^2$

    (b) $E(\frac{1}{X+1}) = \frac{1}{p(n+1)}(1-(1-p)^{n+1})$

    (c) If $E(X) = 6$ and $Var(X) = 2.4$, Find $P(X = 5)$

21. An archer shows an arrow at a target. The distance of the arrows from the centre of the target is a r.v X whose pdf is given by

$$f(x) = \begin{cases} \frac{3+2x-x^2}{9} & x \leq 3 \\ 0 & x > 3 \end{cases}$$

The archers score is determined as follows:

Distance	X < 0.5	0.5 ≤ X < 1	1 ≤ X < 1.5	1.5 ≤ X < 2	X > 2
Score	10	7	4	1	0

Construct the p.m.f for the archer's score, and find the archer's expected score.

22. A discrete r.v has pmf $P(X = n) = \left(\frac{1}{2}\right)^n$. Let

$$Y = \begin{cases} 1 & \text{if } x \text{ is even} \\ -1 & \text{if } x \text{ is odd} \end{cases}$$

Find the expected value of Y

23. A discrete r.v X has pmf of the binomial distributiom with parameters n and p . Let

$$Y = \begin{cases} 1 & \text{if } x \text{ is even} \\ 0 & \text{if } x \text{ is odd} \end{cases}$$

Find the expexted value of Y.

24. Show that the Geometric distribution r.v satisfies the equation

$$P(X > i + j | X > i) = P(X > j)$$

25. A geological study indicates that an exploratory oil well should strike oil with probability 0.2.

    (a) What is the probability that the first strike comes on the third well drilled?

    (b) What assumptions did you make to obtain the answer in (a).

26. At a checkout counter customers arrive at an average of 2 per minute. Find the probability that

    (a) at most 4 will arrive at any given minute.

    (b) at least 3 will arrive during an interval of 2 minutes

    (c) 5 will arrive in an interval of 3 minutes.

27. On a given day, each computer in the lab has at most one crash. There is a 5% chance that a computer has a crash during the day, independent of the performance of any other computers in the lab. Find the probability that on any given day, there are;

    (a) exactly 3 crashes

    (b) at most 3 crashes

    (c) at least 3 crashes

    (d) more than 3 and less than 6.

28. A candidate is taking a multiple-choice exam. For each tested in the exam, there are 5 possible choices i.e A, B, C, D, and E. Because the candidate zero knowledge of the subject, he relies on pure guesswork to answer each question independent of how he answers any previous questions. Find

    (a) the probability that the candidate answers three questions wrong in a row before he finally answers the fourth question correctly.

    (b) let X= the number of problems the candidate answers wrong in a row before he finally guesses a correct answer.

(c) When the candidate's exam is being graded, the grader finds that the candidate has answered the first five exam questions all wrong. Can the grader conclude that the candidate is more or less likely to guess a problem right in other problems.

29. You throw a die repeatedly until you get a 6. What is the probability that you need to throw more than 20 times to get 6?

30. Customers walk in a store at an average rate of 20 per hour. Find the probability that

    (a) no customers have arrived at the store in 10 minutes.

    (b) no more than 4 customers have arrived at the store in 30 minutes.

31. The number of typos in a book has a poisson distribution with an average of 5 typos per 100 pages. What is the probability that you cannot find a typo in 50 pages.

32. A beach resort buy a policy to insure against loss of revenues due to major storms in the summer. The policy pays a total of $ 50,000 if there is only one major storms during the summer, a total of $ 100,000 if there are two major storms, and a total payment of $ 200,000 if there are more than two major storms. The number of major storms in one summer is modeled by a Poisson distribution with mean of 0.5 per summer. Find

    (a) Find the expected premium for this policy during one summer.

    (b) the standard deviation of the cost of providing this insurance for one summer.